

Exhaustive Dataset and Tooling Survey for Semiconductor SEM Navigation-Error Recovery

Solving the Navigation-Error Recovery problem in semiconductor wafer inspection requires locating a fine-resolution reference pattern (1000×1000 pixels) within a search image (1000×1000 pixels) that covers a $10\times$ larger physical field of view (FOV). This $10\times$ linear scale disparity creates a $100\times$ areal coverage ratio difference, introducing critical computer vision challenges:

- High-frequency spatial aliasing and intense pattern repetition in dynamic random-access memory (DRAM) and FinFET architecture arrays.
- Non-Gaussian scanning electron microscope (SEM) noise, secondary electron surface charging artifacts, edge-blooming, and contrast inversion.
- Feature scale mismatches where standard local descriptors (e.g., SIFT, ORB, or uncalibrated CNN feature maps) fail due to spatial resolution quantization differences.

Because commercial metrology vendors like Applied Materials do not distribute proprietary production SEM navigation datasets, algorithms must be trained, validated, and benchmarked through a combination of open SEM image corpora, cross-scale registration benchmarks, physics-based SEM simulators, and procedural computer-aided design (CAD) layout engines. This report provides an exhaustive, curated survey of public datasets, synthetic generators, pretraining corpora, and algorithmic repositories directly applicable to this task.

1. Best Directly Usable Datasets

Datasets in this category provide paired cross-scale imagery, controlled multi-magnification scanning electron micrographs, or explicit scale-disparity matching annotations suitable for training template matching and localization models.

NIST Detection Limits for SEM Image Segmentation Dataset

- **Dataset Name:** NIST Detection Limits for SEM Image Segmentation Dataset¹
- **Direct Download URL:** https://data.nist.gov/od/ds/mds2-3838/intensity_sets.zip [cite: 2]
- **Official Homepage:** <https://www.nist.gov/data-publications/detection-limits-sem-image-segmentation> [cite: 2]
- **Original Paper:** Bajcsy, P., Sathe, P., & Vladar, A. (2025). "Detection Limits for SEM Image Segmentation," *NIST Data Publication / CHIPS Metrology Program*¹.
- **Citation:** Bajcsy, P., Sathe, P., Vladar, A. (2025). Detection Limits for SEM Image Segmentation. National Institute of Standards and Technology. DOI:

10.18434/mds2-3838².

- **License:** Public Domain / NIST Open Data Policy¹.
- **Number of Images:** 6 simulated SEM image collections spanning systematic noise/contrast steps, accompanied by trained TensorFlow models and metric suites¹.
- **Resolution:** High-resolution electron microscopy arrays (1024×1024 and equivalent grids)¹.
- **Image Modality:** SEM Grayscale (Simulated Physics-Based)¹.
- **Data Type:** Semiconductor-like nanostructures, calibration lattices, and dimensional metrology features¹.
- **Annotation Type:** Pixel-level segmentation masks, ground-truth intensity distributions, and AI accuracy metrics¹.
- **Suitability:** Localization, template matching, image registration, feature matching, synthetic generation, pretraining¹.
- **Advantages:** Developed under the NIST CHIPS Metrology Program specifically to evaluate automated inspection algorithms under controlled Poisson noise and contrast degradation¹. Offers ground-truth intensity distributions free of mechanical alignment drift¹.
- **Limitations:** Focuses on single-scale fields per file; requires synthetic cropping/rescaling to construct explicit $10\times$ cross-scale reference/search image pairs.
- **Relevance Score:** 8.5 / 10

Multi-Magnification Ge Nanostructures SEM Dataset

- **Dataset Name:** Dataset for Highly Strained Ge Nanostructures (Multi-Magnification SEM Series)³
- **Direct Download URL:** <https://zenodo.org/records/15082448/files/Figure%201.zip> [cite: 3]
- **Official Homepage:** <https://zenodo.org/records/15082448> [cite: 3]
- **Original Paper:** "Dataset for Highly strained Ge nanostructures and direct bandgap transition induced by femtosecond laser," *Semiconductor Science and Technology*, 2024³.
- **Citation:** Zenodo Repository, DOI: 10.5281/zenodo.15082448³.
- **License:** Open Access / Creative Commons Attribution 4.0 International (CC-BY-4.0)³.
- **Number of Images:** Multiple magnification series across discrete sample parameters³.
- **Resolution:** $1024 \times$ pixels³.
- **Image Modality:** SEM Grayscale (Carl Zeiss AG - SUPRA 40 SEM)³.
- **Data Type:** Nanostructured semiconductor surfaces, laser-etched silicon/germanium nanostructures³.
- **Annotation Type:** Embedded acquisition metadata (scale bar, EHT voltage, exact

magnification tags from $5k\times$ to $50k\times$)³.

- **Suitability:** Localization, template matching, image registration, feature matching, pretraining³.
- **Advantages:** Features real physical SEM micrographs of the exact same physical sample location captured across a $10\times$ zoom gradient ($5k\times$ to $50k\times$), matching the $10\times$ field-of-view ratio requirement³.
- **Limitations:** Sample morphology consists of nanostructures rather than Manhattan DRAM/FinFET layouts.
- **Relevance Score:** 8.0 / 10

OETR Cross-Scale Image Matching Benchmark

- **Dataset Name:** OETR Cross-Scale Image Overlap Benchmark⁴
- **Direct Download URL:** <https://github.com/AbyssGaze/OETR> [cite: 4]
- **Official Homepage:** <https://github.com/AbyssGaze/OETR> [cite: 4]
- **Original Paper:** Chen, H., et al. (2022). "OETR: Overlap Estimation Conditioned on Image Pairs with Transformer," *Proceedings of the AAAI Conference on Artificial Intelligence*⁴.
- **Citation:** Chen, H., et al. (2022). OETR: Overlap Estimation Conditioned on Image Pairs with Transformer. AAAI 2022. [cite: 4]
- **License:** Open Source / MIT License⁴.
- **Number of Images:** Over 10,000 image pairs annotated with cross-scale overlapping regions⁴.
- **Resolution:** 1024×1024 native standard evaluation format⁴.
- **Image Modality:** Grayscale & RGB Multimodal⁴.
- **Data Type:** Paired images exhibiting scale disparities ranging from $2\times$ to $15\times$ zoom differences⁴.
- **Annotation Type:** Bounding boxes enclosing co-visible overlap regions, scale transform ratios, and center coordinates⁴.
- **Suitability:** Localization, template matching, image registration, feature matching⁴.
- **Advantages:** Explicitly formulates the problem of finding a fine-scale reference sub-image within a broad search image under scale, viewpoint, and resolution mismatches using Transformer architectures⁴.
- **Limitations:** Modality consists primarily of aerial and terrestrial surface imagery rather than semiconductor SEM layouts; models require fine-tuning to adapt to nanoscale Poisson noise.
- **Relevance Score:** 8.0 / 10

2. Best SEM Datasets

Scanning electron microscopy datasets provide real-world distributions of secondary electron emissions, edge-blooming artifacts, surface roughness, line-edge roughness (LER), and tool noise essential for robust model training.

Dataset Metric / Property	Carinthia Semiconductor Wafer Dataset	WaferDC SEM Wafer Defect Dataset	Multi-Magnification Ge Nanostructures SEM Dataset
Direct Download URL	https://zenodo.org/records/10715190/files/data.zip [cite: 5]	https://github.com/SpatialAILab/WaferDC [cite: 6]	https://zenodo.org/records/15082448/files/Figure%201.zip [cite: 3]
Official Homepage	https://zenodo.org/records/10715190 [cite: 5]	https://github.com/SpatialAILab/WaferDC [cite: 6]	https://zenodo.org/records/15082448 [cite: 3]
Original Paper / Source	Kofler et al., Zenodo 2024 ⁵	Park et al., EAAI 2025 ⁶	<i>Semiconductor Science & Tech.</i> [cite: 3]
License	CC-BY-4.0 ⁵	CC-BY-NC-ND 4.0 ⁶	CC-BY-4.0 ³
Number of Images	4,591 images ⁵	Multi-thousand real & synthetic SEM set ⁶	Multi-scale magnification series sets ³
Resolution	Standard industrial SEM FOVs ⁵	Variable SEM tile sizes ⁶	1024 × TIFF ³
Image Modality	SEM Grayscale ⁵	SEM Grayscale ⁶	SEM Grayscale (Carl Zeiss SUPRA 40) ³
Data Type	Unstructured wafer production layer ⁵	Structured & long-tailed wafer layers ⁶	Nanostructured semiconductor surfaces ³

Annotation Type	6 defect categories ⁵	Bounding boxes, anomaly masks ⁶	Exact zoom metadata (5kx to 50kx) ³
Suitability	Pretraining, feature matching ⁵	Feature matching, registration, pretraining ⁶	Cross-scale template matching, registration ³
Advantages	Real production SEM micrographs from Infineon fab lines ⁵	Multi-cluster memory bank strategy to decouple scale and background clutter ⁶	Real physical 10× zoom series on identical sample locations ³
Limitations	Unstructured wafer layers ⁵	Primary proprietary SEM set requires running codebase ⁶	Etched nanostructures rather than Manhattan IC patterns ³
Relevance Score	7.5 / 10	8.0 / 10	8.0 / 10

3. Best Wafer Inspection Datasets

Topological wafer map datasets capture macro-level spatial defect patterns across entire silicon wafers, offering structural layout priors and spatial context for semiconductor defect analytics.

WM-811K Wafer Map Dataset

- **Dataset Name:** WM-811K Wafer Map Dataset⁷
- **Direct Download URL:** <https://huggingface.co/datasets/lslattery/wafer-defect-detection> [cite: 7]
- **Official Homepage:** <https://www.kaggle.com/datasets/wafer-defect-map>
- **Original Paper:** Wu, M.-J., Jang, J.-S. R., & Chen, J.-L. (2014). "Wafer Map Failure Pattern Recognition and Similarity Ranking for Large-Scale Data Sets," *IEEE Transactions on Semiconductor Manufacturing*⁷.
- **Citation:** Wu, M.-J., et al. (2014). IEEE TSM, DOI: 10.1109/TSM.2014.2316251⁷.
- **License:** Open Academic Access / Public Domain.
- **Number of Images:** 811,457 wafer maps collected from 46,293 manufacturing lots⁷.
- **Resolution:** Variable die grid dimensions (ranging from 26×26 to 100×100).
- **Image Modality:** Categorical Grid Array / Wafer Topography Maps⁸.

- **Data Type:** Full-wafer functional test result maps⁷.
- **Annotation Type:** 9 failure pattern categories (Center, Donut, Edge-Loc, Edge-Ring, Loc, Random, Scratch, Near-Full, Normal)⁹.
- **Suitability:** Pretraining global spatial pattern priors, layout logic modeling⁹.
- **Advantages:** Industry-standard benchmark for global wafer failure pattern distribution⁹.
- **Limitations:** Abstract macro-level die grids rather than nanoscale SEM intensity imagery.
- **Relevance Score:** 4.0 / 10

MixedWM38 Mixed-Type Wafer Defect Dataset

- **Dataset Name:** MixedWM38 Wafer Defect Pattern Dataset⁸
- **Direct Download URL:** <https://github.com/Junliangwangdhu/WaferMap>
[cite: 8]
- **Official Homepage:**
<https://www.kaggle.com/datasets/co1d7era/mixedtype-wafer-defect-datasets>
[cite: 8]
- **Original Paper:** Wang, J., Xu, C., Yang, Z., Zhang, J., & Li, X. (2020). "Deformable Convolutional Networks for Efficient Mixed-type Wafer Defect Pattern Recognition," *IEEE Transactions on Semiconductor Manufacturing*⁸.
- **Citation:** Wang, J., et al. (2020). IEEE TSM, DOI: 10.1109/TSM.2020.3020985⁸.
- **License:** Educational / Open Source⁸.
- **Number of Images:** ~38,000 wafer maps⁸.
- **Resolution:** 52×52 spatial die arrays⁸.
- **Image Modality:** Synthetic / Augmented Die Maps⁸.
- **Data Type:** Mixed-type semiconductor wafer defect patterns⁸.
- **Annotation Type:** Multi-label defect pattern annotations⁸.
- **Suitability:** Pretraining global pattern priors, synthetic layout modeling⁸.
- **Advantages:** High-quality representations of composite, overlapping manufacturing defects⁸.
- **Limitations:** Discrete macro die maps rather than electron beam micrographs.
- **Relevance Score:** 3.5 / 10

4. Best Microscopy Datasets

Microscopy datasets from materials science and structural biology offer rich background textures, periodic lattice structures, and high Poisson noise distributions that parallel the challenging conditions found in semiconductor SEM processing.

NIST Concrete Microstructure SEM Dataset

- **Dataset Name:** 2D Segmentation of Concrete Samples for Training AI Models (NIST SEM Corpus)¹¹
- **Direct Download URL:** <https://data.nist.gov/od/id/mds2-2155>
[cite: 11]

- **Official Homepage:** <https://www.nist.gov/programs-projects/computational-scanning-electron-microscopy> [cite: 12]
- **Original Paper:** "Scanning electron microscopy (SEM) image segmentation for microstructure analysis of concrete using U-Net," *Automation in Construction*, 2022¹³.
- **Citation:** Bajcsy, P., et al. (2022). *Automation in Construction*, DOI: 10.1016/j.autcon.2022.104602¹⁴.
- **License:** Public Domain / NIST Open Data¹.
- **Number of Images:** 4,594 raw field-of-view SEM images¹³.
- **Resolution:** 2048×1536 pixels.
- **Image Modality:** SEM Grayscale¹¹.
- **Data Type:** Complex heterogeneous materials microstructures¹³.
- **Annotation Type:** Pixel-wise semantic segmentation masks (4 classes)¹¹.
- **Suitability:** Pretraining dense feature extractors, validating texture segmentation under low signal-to-noise ratios¹¹.
- **Advantages:** High annotation density and realistic physics-based electron scattering noise¹¹.
- **Limitations:** Heterogeneous organic geometries rather than strict Manhattan IC layouts.
- **Relevance Score:** 6.0 / 10

EMPIAR Electron Microscopy Archive (EMPIAR-10542 & EMPIAR-10093)

- **Dataset Name:** EMPIAR Public Image Archive (MicroED & Cryo-EM Entries)¹⁵
- **Direct Download URL:** <https://www.ebi.ac.uk/empiar/EMPIAR-10542/> [cite: 16]
- **Official Homepage:** <https://www.ebi.ac.uk/empiar/> [cite: 15, 17]
- **Original Paper:** Bücker, R., et al. (2020). "Serial protein crystallography in an electron microscope," *Nature Communications*¹⁶.
- **Citation:** Bücker, R., et al. *Nat Commun* 11, 996 (2020). DOI: 10.1038/s41467-020-14793-0¹⁶.
- **License:** CC0 / Public Domain.
- **Number of Images:** 100,000+ electron micrographs and diffraction frame series¹⁵.
- **Resolution:** Up to 5760×4092 pixels¹⁷.
- **Image Modality:** Transmission Electron Microscopy (TEM), Cryo-EM, MicroED Grayscale¹⁵.
- **Data Type:** Repetitive crystalline nanostructures and molecular arrays¹⁶.
- **Annotation Type:** Spatial indexing streams, diffraction spot coordinates¹⁵.
- **Suitability:** Pretraining, feature matching on repetitive periodic textures¹⁶.
- **Advantages:** Ground-truth electron scattering optics and Poisson shot noise

distributions¹⁵.

- **Limitations:** Non-semiconductor crystalline lattices.
- **Relevance Score:** 5.5 / 10

5. Best Synthetic Dataset Generators & SEM Simulators

To bridge the gap left by proprietary industrial datasets, open-source procedural generators and physics-based simulators enable the generation of infinite ground-truth image pairs

matching the $10\times$ FOV Navigation-Error Recovery specifications.

Synthetic Data Generation Architectural Workflow

The process for generating paired cross-scale SEM imagery relies on a four-stage synthetic pipeline:

1. **CAD / GDSII Layout Engine (gdsfactory):** Procedurally synthesizes vector geometry for DRAM memory arrays, FinFET transistor clusters, and contact grids²⁰. Exports vector masks at explicit physical scale dimensions.
2. **Scale Disparity Cropping:** Extracts a broad search window ($10\mu\text{m} \times 10\mu\text{m}$) and a fine reference window ($1\mu\text{m} \times 1\mu\text{m}$) from the synthesized CAD layout, explicitly recording the ground-truth center offset (x, y) .
3. **Physics-Based / Neural SEM Simulation Engine (NIST JMONSEL / DDPM SEM Generator):** Simulates electron landing energies (1–5 keV), secondary electron yield (SE2), edge blooming, surface charging, and Poisson shot noise onto both layout masks¹⁰.
4. **Final Output Dataset Pair:** Produces a 1000×1000 pixel Search Image ($10\times$ FOV) and a 1000×1000 pixel Reference Image ($1\times$ FOV) with sub-pixel ground-truth alignment annotations.

Key Simulation Software Tools

NIST JMONSEL Physics-Based SEM Simulator

- **Official Homepage:**
<https://www.nist.gov/programs-projects/computational-scanning-electron-microscopy> [cite: 12]
- **Capabilities:** Physics-based Monte Carlo simulator modeling secondary electron (SE), backscattered electron (BSE), and transmission electron (TE) signals directly from 2D/3D CAD geometry inputs (STEP/GDSII)¹². Models landing energies (1 keV to 30 keV), beam spot diameter, dielectric constants, edge blooming, and Poisson noise¹².
- **Suitability:** Renders physically realistic $10\times$ search images and $1\times$ reference images

from identical CAD sources with sub-pixel ground-truth center coordinates (x, y) .

- **Relevance Score: 9.5 / 10**

DDPM Patch-Based Generative SEM Engine

- **Original Paper:** "Addressing Class Imbalance and Data Limitations in Advanced Node Semiconductor Defect Inspection: A Generative Approach for SEM Images," *arXiv / ML for Semiconductor Manufacturing*, 2024¹⁰.
- **Capabilities:** Uses a Denoising Diffusion Probabilistic Model (DDPM) to synthesize semiconductor SEM images that replicate nanoscale surface roughness, stochastic line-edge roughness (LER), and metrology variations across critical dimensions (CD) and pitches¹⁰.
- **Suitability:** Synthesizing diverse, realistic SEM pattern variations across fine and coarse resolutions¹⁰.
- **Relevance Score: 9.0 / 10**

gdsfactory Procedural IC Layout Engine

- **Official Repository:** <https://github.com/gdsfactory/gdsfactory> [cite: 20]
- **Capabilities:** Open-source Python library for EDA layout synthesis²⁰. Programmatically generates GDSII layout geometries for DRAM memory arrays, FinFET transistor clusters, and custom logic layers, exporting high-resolution multi-tone masks at arbitrary physical scales²⁰.
- **Suitability:** Generating procedural layout geometries as input masks for physics-based SEM engines.
- **Relevance Score: 9.0 / 10**

6. Best GitHub Repositories

The following repositories provide open-source implementations, neural network architectures, and layout engines suitable for developing cross-scale semiconductor registration pipelines:

1. **SpatialAILab/WaferDC**
[cite: 6]
 - **URL:** <https://github.com/SpatialAILab/WaferDC> [cite: 6]
 - **Description:** Official PyTorch implementation of the WaferDC framework (EAAI 2025)⁶. Features a multi-cluster memory bank strategy and SegMix anomaly augmentation designed to decouple scale differences and background clutter in SEM wafer images⁶.
2. **AbyssGaze/OETR**
[cite: 4]
 - **URL:** <https://github.com/AbyssGaze/OETR> [cite: 4]
 - **Description:** Implementation of Overlap Estimation with Transformer (OETR)⁴. Provides a dedicated pipeline for predicting shared spatial bounding boxes and scale ratios

between pairs of images with large scale disparities⁴.

3. **gdsfactory/gdsfactory**

[cite: 20]

- **URL:** <https://github.com/gdsfactory/gdsfactory>
[cite: 20]
- **Description:** Procedural layout generation engine capable of programmatically creating GDSII structures (DRAM grids, FinFET arrays, contacts) and exporting high-resolution multi-scale layout masks²⁰.

4. **Junliangwangdhu/WaferMap**

[cite: 8]

- **URL:** <https://github.com/Junliangwangdhu/WaferMap>
[cite: 8]
- **Description:** Contains code and GAN tools for generating synthetic wafer maps and extracting spatial pattern features⁸.

5. **hellozzk/MMIO**

[cite: 10]

- **URL:** <https://github.com/hellozzk/MMIO>
[cite: 10]
- **Description:** Codebase for multimodal industrial inspection and cross-scale defect localization benchmarks¹⁰.

7. Best Pretraining Datasets

Pretraining vision backbones (e.g., Vision Transformers, ResNets) on large-scale industrial or electron microscopy corpora enables networks to learn robust local feature representations before fine-tuning on semiconductor cross-scale alignment tasks.

- **IMDD-1M & VISION Industrial Datasets:** Over 1,000,000 industrial micrographs and inspection images across 60+ material categories¹⁰. Pretraining on these datasets establishes strong low-level edge and texture representations robust to industrial lighting and noise variations¹⁰.
- **Carinthia SEM Dataset:** Contains 4,591 real industrial wafer SEM images⁵. Provides ideal data for self-supervised pretraining (e.g., Masked Autoencoders, DINOv2) to adapt model backbones to electron beam noise and edge blooming⁵.
- **NIST Concrete SEM Microstructure Corpus:** Provides 4,594 real SEM images with detailed pixel-level annotations¹¹. Useful for pretraining dense feature extraction backbones on complex microstructural topographies¹¹.

8. Datasets Unsuitable for this Problem

Several prominent semiconductor and defect datasets are ill-suited for Navigation-Error Recovery due to fundamental modality or resolution mismatches:

- **WM-811K & MixedWM38 Wafer Maps:** These datasets represent entire silicon wafers as low-resolution discrete categorical matrices (e.g., 26×26 grids indicating pass/fail

dies)⁷. They lack nanoscale electron microscopy features, spatial intensity distributions, and cross-scale zoom information, making them unsuitable for image registration or sub-pixel pattern localization⁸.

- **MVTec AD & VisA Anomaly Benchmarks:** Standard industrial anomaly detection benchmarks contain RGB photography of macro-scale objects (e.g., hazelnut, cable, transistor components). They lack the physics-based Poisson noise, secondary electron edge-blooming artifacts, high-frequency spatial periodicity, and severe $10\times$ cross-scale field-of-view differences present in SEM navigation¹⁰.
- **EMPIAR Biological Single-Particle Datasets (e.g., EMPIAR-10045 Ribosomes):** Cryo-EM datasets of isolated biological macromolecules consist of flexible, non-Manhattan geometries with isotropic projections¹⁹. They lack the rigid, highly repetitive, orthogonal line-and-space patterns characteristic of DRAM and FinFET semiconductor layers.

9. Top 10 Datasets & Tools Ranked for the Navigation-Error Recovery Problem

The following table ranks the top datasets, synthetic generators, and computational frameworks specifically for solving the Navigation-Error Recovery challenge:

Rank	Dataset / Tool Name	Category / Type	Modality	Primary Advantage / Role in Pipeline	Relevance Score
1	NIST JMONSEL Engine [cite: 12]	Physics SEM Simulator	Synthetic SEM	Generates exact ground-truth $10\times$ cross-scale reference/search image pairs from CAD ¹² .	9.5 / 10
2	gdsfactory Engine [cite: 20]	Procedural CAD Renderer	Vector CAD / Mask	Programmatically creates DRAM, FinFET, and	9.0 / 10

				logic layout masks ²⁰ .	
3	DDPM SEM Generator [cite: 10]	Diffusion Neural Engine	Synthetic SEM	Synthesizes realistic nanoscale surface roughness and SEM noise across pitch scales ¹⁰ .	9.0 / 10
4	NIST Detection Limits SEM [cite: 1]	Benchmarking Corpus	Simulated SEM	Provides calibrated Poisson noise and contrast variation sweeps ¹ .	8.5 / 10
5	OETR Benchmark [cite: 4]	Algorithmic Pipeline	Multimodal / Pairs	Transformer framework designed specifically for cross-scale region matching ⁴ .	8.0 / 10
6	WaferDC SEM Dataset [cite: 6]	SEM Defect Corpus	Real SEM	Memory bank architecture for handling variable scales and backgrounds ⁶ .	8.0 / 10
7	Ge	Magnificati	Real SEM	Contains	8.0 / 10

	Nanostructures SEM [cite: 3]	on Series		physical multi-magnification image sets (5kx to 50kx) ³ .	
8	Carinthia SEM Dataset [cite: 5]	Industrial Wafer SEM	Real SEM	Ideal for self-supervised pretraining on real semiconductor SEM noise ⁵ .	7.5 / 10
9	NIST Concrete SEM [cite: 11]	Microstructure Dataset	Real SEM	Enables pretraining of dense feature extractors on complex SEM textures ¹¹ .	6.0 / 10
10	IMDD-1M Corpus [cite: 10]	Pretraining Corpus	Multimodal RGB/SEM	Provides broad pretraining data for learning robust industrial visual representations ¹⁰ .	5.5 / 10

Recommended Implementation Strategy

To implement a complete pipeline using these open-source tools and datasets:

1. **Layout Mask Synthesis:** Use gdsfactory to generate GDSII layouts containing repetitive DRAM arrays, FinFET fins, and contact matrices²⁰. Extract paired crop regions to

construct 1000×1000 pixel search masks ($10\mu\text{m} \times 10\mu\text{m}$ physical FOV) and 1000×1000 pixel reference masks ($1\mu\text{m} \times 1\mu\text{m}$ physical FOV), recording ground-truth center offsets (x, y) .

2. **SEM Image Simulation:** Pass the vector layout masks through NIST JMONSEL or the DDPM SEM Generator to model secondary electron emission, surface charging, edge blooming, and Poisson noise¹⁰.
3. **Backbone Adaptation:** Pretrain a feature extraction backbone (e.g., Vision Transformer or Swin Transformer) on the Carinthia SEM and WaferDC datasets using self-supervised learning (DINOv2) to adapt feature representations to electron beam noise⁵.
4. **Cross-Scale Pattern Localization:** Fine-tune an OETR coarse-to-fine attention network on the synthetic paired dataset⁴. The network matches fine-scale reference features to coarse-scale search features across the $10\times$ zoom factor, using cross-correlation and bounding-box regression to predict sub-pixel center coordinates (x, y) .

Works cited

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2. Detection Limits for SEM Image Segmentation - National Institute of Standards and Technology, <https://www.nist.gov/data-publications/detection-limits-sem-image-segmentation>
3. Dataset for "Highly strained Ge nanostructures and direct bandgap transition induced by femtosecond laser" - Zenodo, <https://zenodo.org/records/15082448>
4. Guide Local Feature Matching by Overlap Estimation - Association for the Advancement of Artificial Intelligence (AAAI), <https://cdn.aaai.org/ojs/19913/19913-13-23926-1-2-20220628.pdf>
5. Carinthia dataset | Zenodo, <https://zenodo.org/records/10715190>
6. GitHub - SpatialAILab/WaferDC: [EAAI 2025] Long-tailed Detection and Classification of Wafer Defects from SEM Images Robust to Diverse Image Backgrounds and Defect Scales, <https://github.com/SpatialAILab/WaferDC>
7. Islattery/wafer-defect-detection · Datasets at Hugging Face, <https://huggingface.co/datasets/islattery/wafer-defect-detection>
8. Mixed-type Wafer Defect Datasets - Kaggle, <https://www.kaggle.com/co1d7era/mixedtype-wafer-defect-datasets/activity>
9. defect-classification · GitHub Topics, <https://github.com/topics/defect-classification>
10. Daily Papers - Hugging Face, <https://huggingface.co/papers?q=Industrial%20Multimodal%20Defect%20Dataset>

11. PDR: 2D Segmentation of Concrete Samples for Training AI Models - NIST Data Repository, <https://data.nist.gov/od/id/mds2-2155>
12. Computational Scanning Electron Microscopy | NIST, <https://www.nist.gov/programs-projects/computational-scanning-electron-microscopy>
13. Approaches to Training Multi-Class Semantic Image Segmentation of Damage in Concrete, <https://pmc.ncbi.nlm.nih.gov/articles/PMC7849179/>
14. Scanning electron microscopy (SEM) image segmentation for microstructure analysis of concrete using U-net convolutional neural network | Request PDF - ResearchGate, https://www.researchgate.net/publication/364151333_Scanning_electron_microscopy_SEM_image_segmentation_for_microstructure_analysis_of_concrete_using_U-net_convolutional_neural_network
15. EMPIAR-10093 Structure of an ion channel in nano disc - EMBL-EBI, <https://www.ebi.ac.uk/empiar/EMPIAR-10093/>
16. EMPIAR-10542 Serial electron diffraction from hen egg-white lysozyme - EMBL-EBI, <https://www.ebi.ac.uk/empiar/EMPIAR-10542/>
17. EMPIAR-11466 Desulfonema ishimotonii Cas7-11 in complex with Csx29 and Csx30, <https://www.ebi.ac.uk/empiar/EMPIAR-11466/>
18. Case Study: End-to-end processing of encapsulated ferritin (EMPIAR-10716), <https://discuss.cryosparc.com/t/case-study-end-to-end-processing-of-encapsulated-ferritin-empiar-10716/15343>
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